

IMY 761 Deliverable 4

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Overview

For this deliverable, you will define the precise design of your research. This includes the main methodological approach, the techniques and instruments with which you will collect data, the sample of participants from which you will collect data, and how you will analyse your gathered data to infer your final results. This will also include a discussion of what you anticipate the outputs/contributions of your study to be, by which aspects those outputs/contributions are limited, and how you will address ethical considerations in carrying out your research.

The structure of this document is based on the structure of the proposal outlined in Chapter 4 of *Research Methods in Information* by Alison Pickard, excluding the literature review.

Practical instructions

Do not create a new document. Fill everything in on this document and then submit it in a .docx format.

Exercise 1: Title and Introduction

Refer to the appropriate section of Research Methods in Information on writing this section. This should address the requirements highlighted in both the “Clearly defined title and introduction” and “statement of aims and objectives, or a hypothesis” sections; you should provide your research questions here, not hypotheses.

Title of your research (5-20 words)

The Effect of Diegetic Versus Non-Diegetic Failure Feedback on Player Frustration and Perceived Competence in a 2D Platformer Game

Introduction (150-200 words)

Failure is a recurring and consequential feature of digital games, and how a game communicates failure to the player is a significant design decision. Failure feedback can be diegetic conveyed through cues embedded in the game world, such as character animation, environmental sound, or spatial change. It can also be non-diegetic, conveyed through interface elements outside the game world, such as HUD overlays, screen flashes, or text prompts. Research has examined how feedback content shapes player motivation and how diegetic interfaces affect immersion more broadly, but little is known about how feedback modality specifically at the point of failure shapes players' emotional and psychological response. Failure moments are when players are most likely to disengage, making the design of failure feedback central to sustaining motivation during play. Framed using the Self-Determination Theory (SDT), the primary research question is: **To what extent** does diegetic, compared with non-diegetic, failure feedback affect players'

frustration and perceived competence within a controlled 2D platformer? A secondary question asks whether this effect differs according to players' self-reported gaming skill level.

Step 2: Methodology + research methods

Based on your current research questions, you should already have a good idea of whether your design is primarily qualitative or quantitative; remember that for this module you have to have at least some aspects of both approaches in your research design to make it more well-rounded and holistic. Refer to the Methodology section in *Research Methods in Information* for the concerns you need to address in your discussion, which includes the research methods you will use in your study; you can refer to Part 2 of *Research Methods in Information* for a list of research methods. You should also read up about a between-subjects versus a within-subjects design and consider which one is most appropriate for your research.

The example below briefly illustrates how one can describe and motivate the methodology and research methods (with references omitted). Your discussion should go into more depth by motivating the choices in the context of your research topic and intended outcomes.

Methodology

The primary methodology for this research is qualitative, since the main goal is to provide an in-depth discussion of factors that affect the experience of using a visualisation application in both PC and VR platforms. A qualitative approach allows for the research outcomes to be guided by rich, descriptive accounts given by participants which more fully describe their experiences and attitudes towards the different platforms (ref). Quantitative data will also be used to provide a baseline overview of the differences between the two platforms in terms of [measures to be used as determined by research questions] (ref). This will also allow for a degree of triangulation between the two sources of data (ref).

Research methods

The main method to be used is that of experimental research, since there is a test condition in the form of a VR application and a control condition in the form of a PC application (ref). This design allows for a degree of causality to be determined as a result of using the VR version, since the results can be compared with that of the PC version (ref). This means that half of the participants will use the VR version and half will use the PC version and participants will be assigned to their group by...

Exercise 2

Discuss the methodology and research methods you will use and motivate it with appropriate references. You should include other sources in addition to *Research Methods in Information*, such as more focused papers on the research methods you plan to use.

Methodology (100-200 words)

This research adopts a mixed-methods design combining a small quantitative component with a qualitative one. Its central aim is to explore whether failure feedback modality is associated with differences in players' frustration and perceived competence, and to understand how players make sense of failure under each condition. A quantitative element is appropriate because frustration and perceived competence can be captured with validated questionnaires, allowing the two experimental conditions to be compared (Pickard, 2013), consistent with Self-Determination Theory's treatment of competence and frustration as measurable need states (Ryan and Deci, 2000). Given the small sample, these quantitative results are treated as descriptive and indicative rather than as a basis for statistical inference. The qualitative component brief post-play interviews therefore carries the primary interpretive weight, letting participants describe in their own words how the feedback shaped their emotional response and sense of capability. Throughout, the two conditions differ only in the channel through which failure is communicated, not its informational content, which isolates diegesis from informativeness and strengthens the validity of the comparison (Creswell, 2014).

Research methods (200-400 words)

The primary method is experimental research, in which the independent variable failure feedback modality is manipulated across two conditions, diegetic and non-diegetic, while all other aspects of the game are held constant (Pickard, 2013). Because the conditions are matched on informational content, level design, and difficulty, any difference in frustration or perceived competence can be attributed to feedback modality rather than to differences in challenge or interface quality (Creswell, 2014).

A between-subjects design is used, with each participant experiencing only one condition. This is preferable to a within-subjects design because exposure to both conditions would introduce learning effects and comparison bias, since a participant who completed one version would approach the second already familiar with the levels (Field, 2018). Between 10 and 15 participants will be recruited and assigned to conditions using randomisation stratified by self-reported gaming skill, so that more and less experienced players are spread across both groups rather than clustering in one. At this sample size the study is exploratory: it is not powered for inferential statistical testing, so the quantitative findings are treated as descriptive and the qualitative interviews carry the main interpretive load. Data are collected through two techniques.

A post-play questionnaire measures frustration and perceived competence using validated instruments from the relevant subscales of the Basic Needs in Games Scale (BANGS) (Ballou et al., 2024), and frustration items from the Game Experience Questionnaire (IJsselsteijn et al., 2013) administered immediately after play. A short semi-structured interview then explores how participants experienced failure and what shaped their motivation to continue. Questionnaire responses are summarised descriptively (i.e. means, standard deviations, and effect-size estimates), while interviews are transcribed and analysed using Braun and Clarke's (2006) six-phase thematic analysis to explain the patterns the questionnaire data suggest.

Step 3: Data collection techniques + research instruments

Discuss the data collection techniques you will use to collect your data; refer to Part 3 of *Research Methods in Information* for a list of data collection techniques. An important consideration here is the appropriateness of the specific data collection techniques for the data you intend to collect. E.g., while it may be easier to use open-ended questionnaires to collect open-ended data, participants generally find it easier to express themselves verbally, which might make interviews a more appropriate option. However, if you are comparing different versions of something and you want holistic data on the difference, you might need to expose participants to both versions to allow them to discuss the differences, which could motivate a within-subjects approach.

If you are doing some type of subjective self-report, i.e., some type of measure where participants themselves give scores/ratings, you need to find a validated instrument for the measure you are gathering data on. E.g., if you are planning on measuring the user's experience of feeling "present" in a virtual environment, you might use the Witmer & Singer presence questionnaire (Witmer and Singer, 1998), the ITC-SOPI (Lessiter et al., 2001), or the Slater-Usoh-Steed presence questionnaire (Slater et al., 1994) as these are all validated questionnaires for this purpose. It is important that you use an existing questionnaire that has been validated for your purpose as this lends ecological validity to your results.

You can find validated questionnaires on Google Scholar by searching for "validated questionnaire" and the measure you are interested in. Make sure that you can also find the actual questionnaire and, if applicable, guide on using it correctly. Also make sure to motivate the use of a specific questionnaire that is clearly in line with the specific outcomes of your study. For example, simply stating that you will be using the user experience questionnaire (UEQ) because your study focuses on user experience is insufficient, in this case you would need to discuss what this questionnaire includes as outcomes and why these specific outcomes (as opposed to that of other UX questionnaires) are relevant.

For this section, you will also need to collect and/or create all the research instruments you plan to use, e.g., validated questionnaires, interview questions, etc.

The example below briefly illustrates how one can describe and motivate the methodology and research methods (with references omitted). Your discussion should go into more depth by motivating the choices in the context of your research topic and intended outcomes.

Data collection techniques

The two data collection techniques to be used are (quantitative) questionnaires and (qualitative) interviews. The numerical data provided by the questionnaire will allow for a direct comparison between the two application versions based on [measures to be used] (ref). After using whichever version a participant has been assigned to, they will be asked to fill out the questionnaire based on their experience of using the application. When they are done, an interview will be conducted to further gauge their experience... (ref).

Research instruments

The user experience questionnaire (UEQ) (Schrepp et al., 2017) will be used to collect hedonic and pragmatic user experience data. This questionnaire was chosen because... The interview questions will focus on... Both the UEQ and interview questions are included below.

Exercise 3

Discuss the data collection techniques and research instruments you will use and motivate it with appropriate references. Add the research instruments you will use (i.e., the questionnaires, interviews, etc.) as an appendix to this document.

Data collection techniques (200-400 words)

This study uses two complementary data collection techniques: a post-play questionnaire and a short semi-structured interview. Each is matched to the kind of data it best captures, and together they provide a quantitative measure of players' responses and a qualitative account of the reasoning behind those responses (Pickard, 2013).

A questionnaire is the appropriate technique for the two outcomes, "frustration" and "perceived competence", because these are internal psychological states that only the participant can report, and validated instruments measure them on a numerical scale that allows the two conditions to be compared (Field, 2018). It is administered immediately after gameplay, while the experience is fresh, using the response format of the validated scales from which its items are drawn. Using established questionnaires rather than self-written items measures ecological validity and makes the results comparable with prior research (Pickard, 2013).

A semi-structured interview is included because questionnaire scores capture the strength of a response but not the reasoning behind it, and participants generally express such reasoning more easily in speech than in writing (Pickard, 2013). A within-subjects approach, which would let participants compare both versions directly, was rejected because exposure to both conditions would introduce learning and comparison effects that confound the outcomes (Field, 2018); each participant therefore plays and discusses only one condition. The interview, lasting roughly five minutes, probes how players experienced failure, what made them want to continue or stop, and how the feedback affected their sense of capability. Interviews are audio-recorded and transcribed for thematic analysis.

Research instruments (200-400 words)

The study uses two existing, validated questionnaires rather than researcher-written items, since established instruments carry demonstrated reliability and validity and make the findings comparable with prior work (Pickard, 2013).

Perceived competence and competence frustration are measured using the relevant subscales of the Basic Needs in Games Scale (BANGS; Ballou et al., 2024). BANGS is chosen over more general need-satisfaction or user-experience questionnaires because it is purpose-built and validated specifically for video games, and because it treats competence satisfaction and competence frustration as distinct constructs rather than as

two ends of a single dimension. This distinction matters here, because the study's interest is specifically whether players feel capable after failure - not their overall enjoyment - a precision that broader scales conflating competence with satisfaction or performance would lose.

Frustration is measured using the relevant items of the Game Experience Questionnaire (GEQ; IJsselstein et al., 2013), specifically its tension/annoyance and negative-affect components. The GEQ is a widely used, validated instrument for assessing player experience, and its frustration-related items provide an established measure of the study's primary emotional outcome that complements the competence-frustration items in BANGS.

Both instruments use a Likert response format and are combined into a single questionnaire to minimise participant burden, with items reproduced exactly from the published scales rather than reworded. A short demographic section collected beforehand records age, gender, and gaming frequency, together with a self-rated skill item used for stratified assignment.

The semi-structured interview schedule is researcher-developed, as is standard for qualitative interviewing, and consists of a small set of open questions with scope for follow-up, designed to elicit the reasoning behind the questionnaire responses. The validated questionnaire and the interview schedule appear in Appendix B.

Step 4: Sampling + analysis

Discuss your sampling strategy and your method of analysis. Your approach to sampling should make sense within the context of your study and should be linked to your intended research outcomes; you should aim to include at least 10 participants in your data collection. You should also include your intended number of participants and consider the distribution of participants, i.e., between- or within-subjects, according to this.

Also discuss your plan for analysing the data. Start by describing how you will process your data, e.g., transcribing interviews, retrieving text data (from whichever tools you used for collecting data), etc. Then describe how you will analyse your data for the purpose of answering your research questions. For quantitative data, this would include the statistical methods you plan to use and for qualitative data this would be the specific approach to analysing data from interviews, observations, etc. Again, you should refer to more specific resources for these approaches, e.g., thematic analysis, grounded theory, etc.

Exercise 4

Discuss your strategies for sampling and analysis and motivate it with appropriate references.

Sampling (100-200 words)

The study will recruit between 10 and 15 participants through non-probability convenience

sampling using **university students**. Convenience sampling of this kind is appropriate for a small, exploratory module project: it makes recruitment and scheduling feasible within the data-collection window, and the study's aim is to surface indicative patterns and rich participant accounts rather than statistically generalisable estimates (Pickard, 2013).

A between-subjects distribution is used, with participants split as evenly as possible between the diegetic and non-diegetic conditions (roughly five to eight per condition). Each participant experiences only one version, avoiding the learning and comparison effects that exposure to both would introduce, since familiarity with the levels would carry over and confound the outcomes (Field, 2018). To keep gaming skill from clustering in one condition, participants self-report their skill level and are then assigned using randomisation stratified by that rating. Given the sample size, the quantitative data is treated as descriptive and the interviews as the primary source of insight.

Analysis techniques (100-300 words)

Data processing precedes analysis. Questionnaire responses will be scored according to each instrument's published scoring guide and compiled into a single dataset by participant code. Interview recordings will be transcribed verbatim and anonymised before analysis.

The quantitative analysis is descriptive rather than inferential, in keeping with the small sample. The two conditions will be compared on frustration and perceived competence using group means, standard deviations, and effect-size estimates, presented alongside individual-level responses so that patterns and outliers remain visible (Field, 2018). No significance testing is performed, since the sample cannot support reliable statistical inference and any differences observed between conditions are reported as indicative trends to be examined in future, adequately powered research.

The qualitative analysis carries the primary interpretive weight. Interview transcripts will be analysed using Braun and Clarke's (2006) six-phase reflexive thematic analysis: familiarisation with the data, generating initial codes, searching for themes, reviewing themes, defining and naming them, and producing the report. This approach suits a small, information-rich dataset and allows the study to explain how players make sense of failure under each condition that compensates for the limited statistical reach of the quantitative measures, bringing a triangulated account of how feedback modality shaped players' experience.

Step 5: Study outcomes, limitations, and considerations

This last step focuses on contextualising the study as a whole by discussing what the intended outcomes are, but also the limitations within which these need to be considered. We will also give an overview of the ethical considerations that play a role in the design of the research. Start by describing the anticipated outputs of the study, which should directly follow from the description of the data to be collected and analysed. Highlight what gap is

being addressed in the body of knowledge on the topic, how the outputs from your study address this gap, and why this matters.

Also discuss the study's limitations. You should be transparent about the factors that limit the extent to which the anticipated can be understood and applied more broadly, based on the design choices made for the study itself, as well as anticipated logistical issues that could affect your study. Finally, address the ethical considerations that arise as a result of using people in your study. The most important issues to address are 1) informed consent, 2) right to withdraw, and 3) anonymity. You also need to include the forms you will provide participants with, which informs them of all these factors.

Exercise 5

Discuss your strategies for sampling and analysis and motivate it with appropriate references. Include the informed consent forms as an appendix to this document. Also draw up a timeline with all the main deadlines as well as intermediary goals based on your study requirements.

Anticipated outputs (50-150 words)

The study's primary output is a descriptive comparison of how diegetic versus non-diegetic failure feedback relates to players' frustration and perceived competence, complemented by a qualitative account, from thematic analysis, of how players interpret and respond to failure under each condition. Together these indicate whether feedback modality is a promising avenue for further, larger-scale investigation. The work addresses a clear gap: while prior research has examined feedback content and the broad effect of diegetic interfaces on immersion (Iacovides et al., 2015), no study has isolated feedback modality at the point of failure as an experimental manipulation with frustration and perceived competence as outcomes. This matters because failure is where players most often disengage, making its presentation central to sustaining motivation during play.

Limitations (100-200 words)

The most significant limitation is sample size. With 10 to 15 participants, the study is exploratory and its quantitative findings are descriptive rather than statistically generalisable, as it cannot reliably detect effects, and any differences observed between conditions must be read as indicative rather than conclusive (Field, 2018). Recruiting from university students further restricts diversity and may introduce bias, since participants may share characteristics. Player skill is treated descriptively rather than as a tested moderator, despite evidence that its influence may be considerable (Iacovides et al., 2015). The study is also confined to a single custom 2D platformer, limiting generalisation to other genres or longer sessions. Because frustration and competence are measured solely through self-report, the findings depend on participants reporting their states accurately. Finally, content-matching the two conditions requires the diegetic demonstration to be fairly explicit, which a stricter reading could view as instructive feedback rather than naturalistic in-world cueing, and the pilot manipulation check mitigates but does not eliminate this risk.

Ethical considerations (100-200 words)

Because the study involves human participants and deliberately induces mild frustration through repeated in-game failure, several safeguards are central. First, informed consent: before participating, each person receives a participant information sheet describing the study's purpose, procedure, duration, and the fact that gameplay is intentionally challenging, and provides written consent (see Appendix A). Second, the right to withdraw: participants are told they may stop at any point, decline any interview question, and withdraw their data afterward without penalty or explanation. Third, anonymity: no directly identifying information is recorded; each participant is assigned a code, questionnaire responses are stored against that code only, and interview recordings are transcribed, anonymised, and the originals deleted after transcription. Because participants are recruited from university campus, particular care is taken to avoid perceived pressure: participants are reminded that taking part is genuinely optional and that declining or withdrawing carries no social cost. Data is stored securely and used solely for this research (Pickard, 2013). A short debrief after each session explains that the failure was designed deliberately.

Timeline:

Phase	Deadline	Intermediary goals
System Design	11 May	Completed
Ethics clearance	early July	Submit protocol and consent forms (Appendix A) to the institutional ethics committee; approval needed before September data collection and may take several weeks
Beta build	June–August	Build Godot 4 prototype: single FailureEvent pipeline, companion entity present from level 1, diegetic and non-diegetic presenters, checkpoint respawn
System Beta	11 August	Functionally complete build submitted for supervisor feedback
Pilot + manipulation check	Mid–late August	Run 4–6 pilot players; confirm informativeness parity between conditions; tune diegetic demonstration until ratings match; fix bugs from feedback
System Final	14 September	Final, data-collection-ready build demonstrated

Data collection	September	Recruit 10–15 participants from university campus; stratify by self-reported skill; run sessions (gameplay → questionnaire → interview)
Data processing + analysis	Late September–October	Transcribe and anonymise interviews; compile questionnaire responses; descriptive statistics; reflexive thematic analysis
Results & Discussion	26 October	Present descriptive results and themes; interpret against the research questions
Final Article	9 November	Compile summarised deliverables into the formatted article
Incorporate feedback	November	Revise article on supervisor/marker feedback
Exam Submission	23 November	Submit final version of the article

Appendices

Appendix A: Participant Information, Consent, and Debrief

A.1 Participant Information Sheet

Study title: The Effect of Diegetic Versus Non-Diegetic Failure Feedback on Player Frustration and Perceived Competence in a 2D Platformer Game

Researchers: Thato Kalagobe and Chenoa Perkett, IMY 761

You are invited to take part in a research study. Please read this information before deciding whether to participate.

Purpose: This study investigates how different ways of communicating failure in a video game affect players' experience. You will play a short 2D platformer, complete a brief questionnaire, and take part in a short interview.

What participation involves: One session of approximately 25–30 minutes, made up of gameplay (up to 15 minutes), a short questionnaire, and a roughly five-minute audio-recorded interview. The game is designed to be challenging, and you may fail and retry sections. This is a normal and intended part of the study.

Risks: Risks are minimal and limited to mild frustration arising from in-game challenge, comparable to playing an ordinary video game.

Benefits: There are no direct benefits to you. Your participation will contribute to research on game design and player experience.

Voluntary participation: Taking part is entirely your choice. Because you may know one of the researchers, please understand that there is no expectation or pressure to take part, and declining will not affect your relationship with the researchers in any way.

Contact: For questions about the study, contact [researcher name, email]. For concerns about your rights as a participant, contact [ethics committee, email].

A.2 Informed Consent Form

Please initial each box to confirm, then sign below.

- ☐ I have read and understood the participant information sheet.
- ☐ I understand what my participation involves, including that the game is intentionally challenging.
- ☐ I understand that my participation is voluntary.
- ☐ I understand that I may withdraw at any time, and may withdraw my data after the session, without giving a reason and without penalty.
- ☐ I consent to my interview being audio-recorded.

☐ I understand that my data will be kept anonymous and stored securely.

☐ I agree to take part in this study.

Participant name: _____ Signature: _____ Date: _____

Researcher name: _____ Signature: _____ Date: _____

A.3 Right to Withdraw Statement

Your participation is completely voluntary. You may stop the game at any point, skip any questionnaire item or interview question, and end the session whenever you wish, without giving a reason and without any negative consequence. You may also withdraw your data after the session by contacting the researchers, in which case all of your responses and recordings will be permanently deleted. Withdrawal carries no penalty and will not affect your relationship with the researchers.

A.4 Data and Anonymity Statement

No directly identifying information (such as your name or contact details) will be linked to your responses. On arrival you will be assigned a participant code, and your questionnaire responses will be stored against that code only. Interview audio will be transcribed, anonymised (with any identifying details removed), and the original recordings deleted after transcription. All data will be stored securely in password-protected, access-restricted storage, accessible only to the researchers and supervisor, and used solely for this research project and its academic reporting. Anonymised data may be retained for the duration required by the institution and then deleted.

A.5 Debrief Note

Thank you for taking part. This study compared two ways of communicating failure in the game: one that delivers feedback through the game world (a companion character that demonstrates the correct path) and one that delivers the same information through on-screen interface elements (such as text and arrows). Both versions gave the same information; only the way it was delivered differed. The repeated failure you experienced was deliberately designed as part of the study and is not a reflection of your ability. We are interested in how the style of feedback affected your frustration and your sense of competence. If you have any questions, or wish to withdraw your data, please let us know.

Appendix B: Research Instruments

B.1 Demographic questionnaire (completed before gameplay)

1. Age: _____
2. Gender: _____
3. How often do you play video games? (Never / Rarely / Monthly / Weekly / Daily)
4. How would you rate your overall gaming skill? (1 = Beginner to 7 = Expert): _____

B.2 Post-play questionnaire (validated instruments)

This questionnaire combines validated subscales from two published instruments: the Basic Needs in Games Scale (BANGS; Ballou et al., 2024) for competence satisfaction and competence frustration, and the Game Experience Questionnaire (GEQ; IJsselsteijn et al., 2013) for tension/frustration. The BANGS items use a 7-point scale (1 = Strongly disagree to 7 = Strongly agree); the GEQ items use the GEQ's own 5-point format (0 = Not at all to 4 = Extremely). Present all items in a mixed order; they are grouped here by source for clarity.

BANGS - Competence satisfaction:

1. I felt capable while playing.
2. I felt competent at the game.
3. I successfully completed challenges in the game.

BANGS - Competence frustration:

4. I felt incapable while playing.
5. I felt like a failure when I played.
6. The game made me doubt my ability.

GEQ - Tension/Annoyance :

7. I felt annoyed.
8. I felt irritable.
9. I felt frustrated.

B.3 Semi-structured interview schedule

Conducted immediately after the questionnaire, lasting approximately five minutes, audio-recorded. The opening question eases the participant in; the core questions map directly onto the study's outcomes (frustration, perceived competence, motivation to continue). Follow-up prompts are used as needed.

Opening:

1. How did you find the game overall?

Core:

2. How did you feel when you failed during the game?
3. When the game responded to your failure, **what did you think had gone wrong?**
4. Did the feedback after failing make you feel more or less capable of improving? Why?
5. Did the feedback make you want to keep playing, or want to stop? What was it about the feedback that did that?
6. Would you describe the feedback as clear, helpful, distracting, or frustrating — and why?
7. After failing **several** times, how did the feedback affect your confidence?

Closing:

8. Is there anything else about your experience you'd like to add?

Follow-up prompts : "Can you say a bit more about that?" / "What do you mean by...?" / "Why do you think you felt that way?"

References

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